

JEE MAIN REVISION MASTER: INDEFINITE INTEGRATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Standard Integrals, Substitution & By Parts	25 PYQs	Integration Techniques
Moderate	Partial Fractions & Special Algebraic Integrals	3 PYQs	Algebraic Integrals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Standard Integrals, Substitution & By Parts	$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$, $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right + C$, $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$
Partial Fractions & Special Algebraic Integrals	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$, $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln x + \sqrt{x^2 \pm a^2} + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Standard Integrals, Substitution & By Parts (Very High Yield — 25 PYQs)

Core Principle: Fundamental integrals, substitution methods, ILATE integration by parts, and linear/quadratic partial fractions.

■ **Caution / JEE Pitfall:** In integration by parts, follow the ILATE priority strictly for choosing the first function u (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential).

■ **Speed Trick:** The pattern $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ appears with high frequency in both JEE Main and KCET.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If 'n' is a natural number, then $\int \frac{\sin^n x \cos^{n+2} x dx}{\dots}$ " [Ans: D]
- [2025 Shift] Q#3 (MCQ): "The value of $\int \frac{dx}{(x+1)(x+2)}$ is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): " $\int \frac{dx}{x^2(x^4+1)^{3/4}}$ equals..." [Ans: N/A]

3.2 Partial Fractions & Special Algebraic Integrals (Moderate Yield — 3 PYQs)

Core Principle: Integration using partial fraction decompositions, linear over quadratic forms, and standard quadratic root substitutions.

■ **Caution / JEE Pitfall:** Check if degree of numerator $\geq$ degree of denominator; if so, polynomial long division **MUST** be performed first.

■ **Speed Trick:** For integrals of type $\int (dx / a + b \cos^2 x)$, divide numerator and denominator by $\cos^2 x$ and substitute $\tan x = t$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int e^{-x} \log 2 \cdot 2^x dx = \dots$ " [Ans: B]
- [2021 Shift] Q#3 (MCQ): "The value of $\int \frac{e^x dx}{(1+x)^2}$ is equal to..." [Ans: N/A]
- [2021 Shift] Q#4 (MCQ): "The value of $\int e^x [(1+\sin x) / (1+\cos x)] dx$ is equal to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\int e^{-x} \log 2 \cdot 2^x dx =$$

- (A) $\log x + C$
- (B) $x + C$ [Correct]
- (C) $(1/x) + C$
- (D) $\frac{1}{x^2} + C$

Correct Answer: B

Solution Analysis: Since 2^x can be written as $2^x = e^{x \log 2}$ the integrand becomes $e^{-x} \log 2 \cdot 2^x = e^{-x} \log 2 \cdot e^{x \log 2}$. Using $e^a \cdot e^b = e^{a+b}$, $e^{-x} \log 2 + x \log 2 = e^0 = 1$. So the integral is $\int 1 dx = x + C$. So the correct option is $x + C$ Option B.

Benchmark Problem 2: KCET 2026 (Q#2)

If 'n' is a natural number, then $\int \frac{\sin^n x \cos^{n+2} x}{1} dx =$

- (A) $\frac{\tan^{n-1} x}{n-1} + C$
- (B) $\frac{\tan^n x}{n} + C$
- (C) $\frac{\tan^{n+2} x}{n+2} + C$
- (D) $\frac{\tan^{n+1} x}{n+1} + C$ [Correct]

Correct Answer: D

Solution Analysis: Given, $\int \frac{\sin^n x \cos^{n+2} x}{1} dx$. We can rewrite the integrand by separating powers of $\cos x$: $\frac{\sin^n x \cos^{n+2} x}{1} = \frac{\sin^n x \cos^n x}{1} \cdot \frac{1 \cos^2 x}{1}$. Now, $(\sin x / \cos x) = \tan x$ and $\frac{1 \cos^2 x}{1} = \sec^2 x$. So, $\frac{\sin^n x \cos^{n+2} x}{1} = \tan^n x \sec^2 x$. Hence, $\int \frac{\sin^n x \cos^{n+2} x}{1} dx = \int \tan^n x \sec^2 x dx$. Put $t = \tan x$. Then, $dt = \sec^2 x dx$. So the integral becomes $\int t^n dt = \frac{t^{n+1}}{n+1} + C$. Putting back $t = \tan x$, $\int \frac{\sin^n x \cos^{n+2} x}{1} dx = \frac{\tan^{n+1} x}{n+1} + C$. So the correct option is (Option D) $\frac{t}{n+1}$.